

TRADER PERFORMANCE vs. Bitcoin Market Sentiment

An Exploratory Data Analysis & Predictive Modeling Report

Dataset window: May 1, 2023 – May 1, 2025
2,644 days of sentiment data | 211,224 trade-level records | 32 accounts

1. Executive Summary

This report examines whether Bitcoin market sentiment (Fear, Neutral, or Greed, as measured by the Bitcoin Fear & Greed Index) is associated with how retail traders in this dataset actually perform — their win rates, profit and loss, trade sizing, and long/short positioning. The analysis combines descriptive statistics, statistical hypothesis testing, and two families of predictive models (classification and regression) across 211,224 individual trades and 2,644 days of sentiment history.

The core takeaway: sentiment shifts trading behavior far more than it shifts trading outcomes. Traders are demonstrably more active and trade larger sizes during Fear, and they lean short during Greed and long during Fear — but win rates and closed PnL are statistically indistinguishable across sentiment regimes. Individual trader skill (rolling win rate) dominates every other factor in predicting whether a given trade wins or loses.

84.4% Win Rate — Fear	82.4% Win Rate — Greed	82.3% Win Rate — Neutral
---------------------------------	----------------------------------	------------------------------------

2. Dataset & Methodology

2.1 Data Sources

- Sentiment data: Bitcoin Fear & Greed Index, daily values from 2018 to 2025 (2,644 days), later restricted to the overlap window with trade data.
- Trade data: 211,224 individual closed/executed trades across 32 unique trading accounts.
- Overlap window used for all analysis: May 1, 2023 – May 1, 2025.

2.2 Analytical Approach

- Descriptive comparison of trading activity, PnL, position sizing, and long/short bias across the three sentiment buckets (Fear / Neutral / Greed).
- Statistical testing: Mann-Whitney U test (Fear vs. Greed PnL), Pearson correlation (sentiment value vs. daily PnL), and chi-square test of independence (long/short bias vs. sentiment).
- Predictive modeling: Logistic Regression and Random Forest for win/loss classification; Linear Regression and Random Forest for closed PnL regression.
- Feature importance analysis to identify which variables actually drive trade outcomes.

3. Market Sentiment Backdrop

The Fear & Greed Index oscillated across the full 2018–2025 history, cycling repeatedly between extreme fear (sub-20) and extreme greed (above 80). This volatility in sentiment provides a reasonably balanced backdrop against which to test whether trader behavior and performance shift with the crowd's mood.

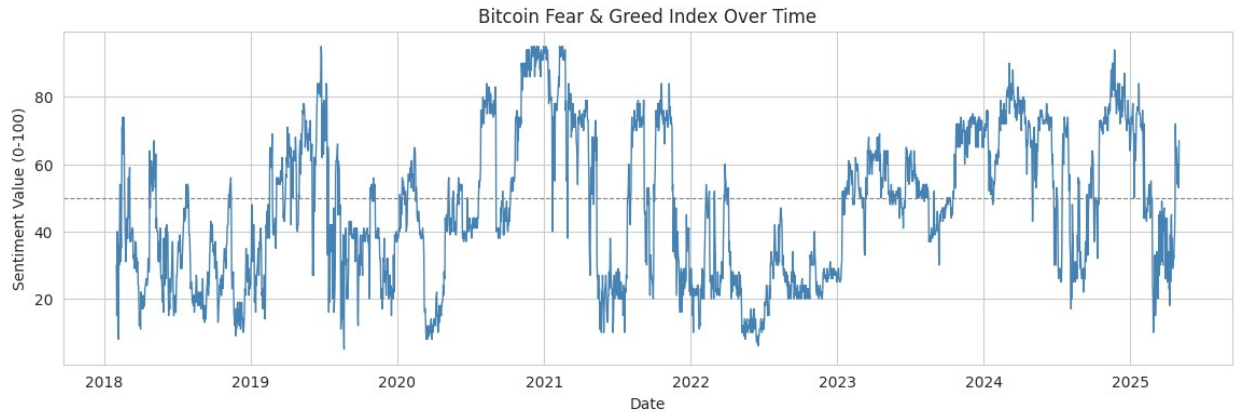


Figure 1. Bitcoin Fear & Greed Index, 2018–2025. Dashed line marks the neutral midpoint (50).

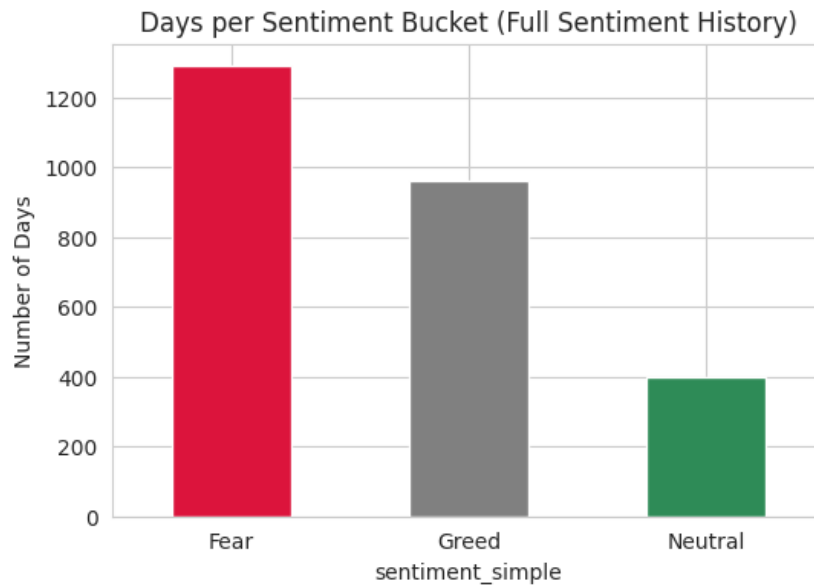


Figure 2. Number of days classified into each sentiment bucket across the full history — Fear is the most common regime, followed by Greed, then Neutral.

4. Trading Activity by Sentiment

Sentiment is clearly linked to how much traders trade — even if not to how well they trade. Activity, volume, and the number of active accounts all peak during Fear and fall to their lowest during Greed.

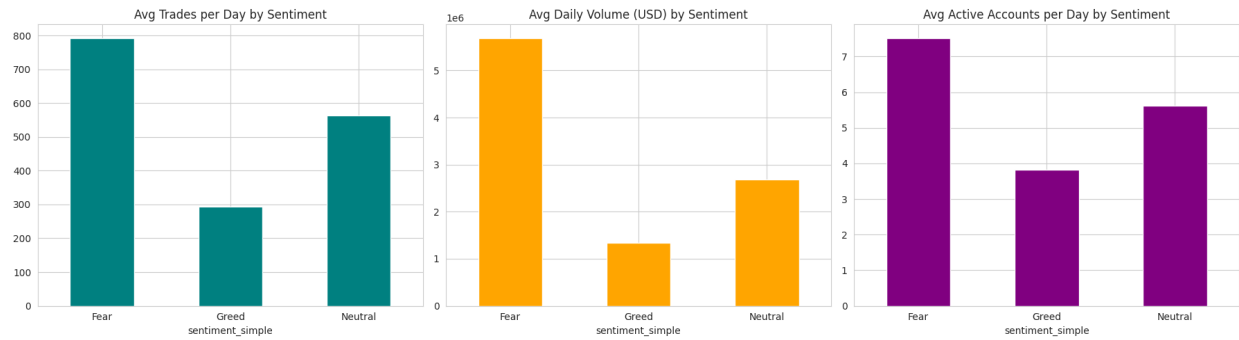


Figure 3. Average trades per day, average daily volume (USD), and average active accounts per day, split by sentiment regime.

- Average trades per day is highest during Fear (~790), roughly 2.7x the Greed level (~295).
- Average daily volume follows the same pattern — Fear sees roughly 4x the volume of Greed.
- More accounts are active per day during Fear (~7.5) than during Greed (~3.8), suggesting fear-driven periods draw in more participants, not fewer.

5. Position Sizing & Directional Bias

5.1 Trade Notional Size

Median trade size is somewhat larger during Fear than during Neutral or Greed conditions, and the spread (upper quartile / whisker) is also wider — consistent with traders taking bigger swings when the market is fearful.

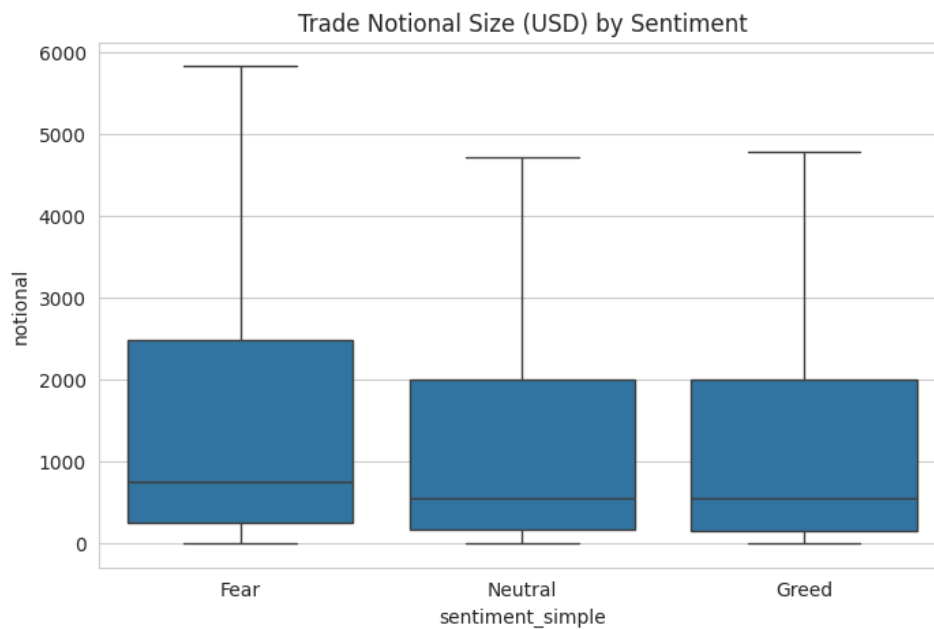


Figure 4. Trade notional size (USD) distribution by sentiment (outliers trimmed for readability).

5.2 Long vs. Short Positioning

Directional bias shows the clearest sentiment relationship in the entire dataset. Long positioning is highest during Fear (58%) and lowest during Greed (33%), where short exposure dominates instead. This is statistically significant: a chi-square test of independence between sentiment and side gives $p < 0.00001$, with a moderate effect size (Cramér's $V = 0.18$).

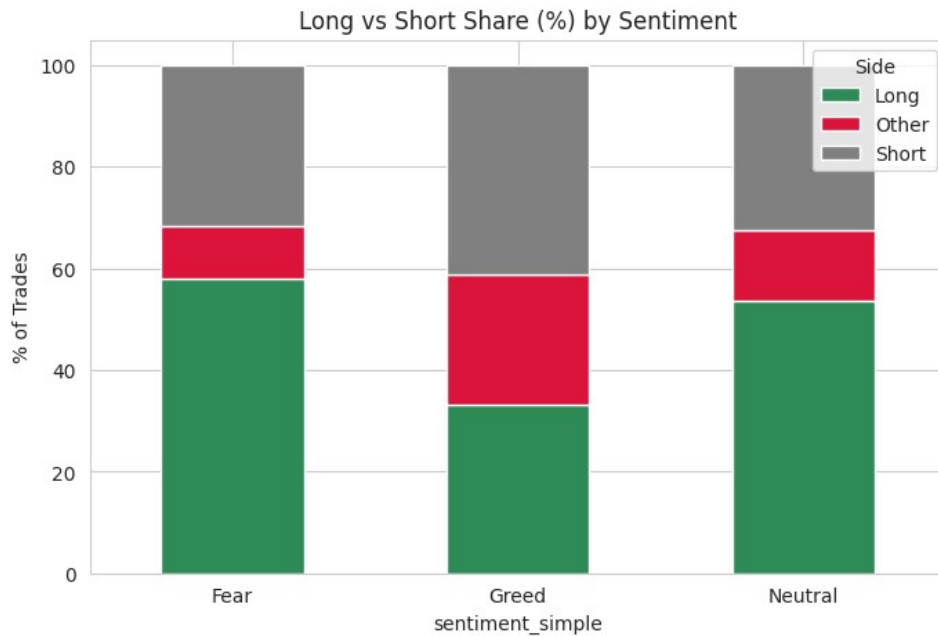


Figure 5. Share of trades that are Long, Short, or Other, by sentiment bucket.

This pattern is intuitive — traders lean long (buying the dip) when sentiment is fearful and lean short (fading the rally / taking profit) when sentiment is greedy — a mild contrarian tilt in aggregate trader behavior.

6. Profitability by Sentiment

6.1 Win Rate

Win rate is highest during Fear (84.4%) and lowest during Neutral (82.3%), with Greed in between (82.4%) — but the gap between all three is under 2.1 percentage points, a small margin against a dataset this large.

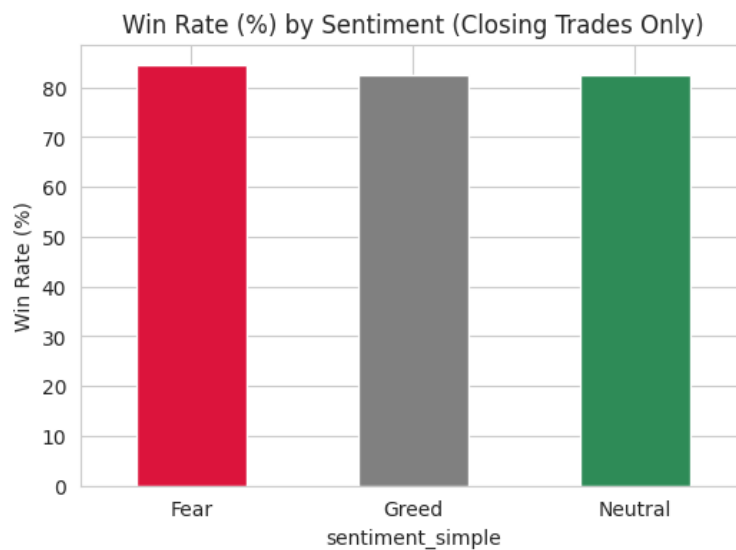


Figure 6. Overall win rate by sentiment bucket (closing trades only).

Breaking sentiment into its full five-way classification (Extreme Fear → Extreme Greed) shows win rate is not monotonic: Extreme Greed (89%) and Fear (87%) post the highest win rates, while Extreme Fear (76%) and Greed (77%) are lowest — indicating the relationship is noisier at the extremes rather than a clean linear trend.

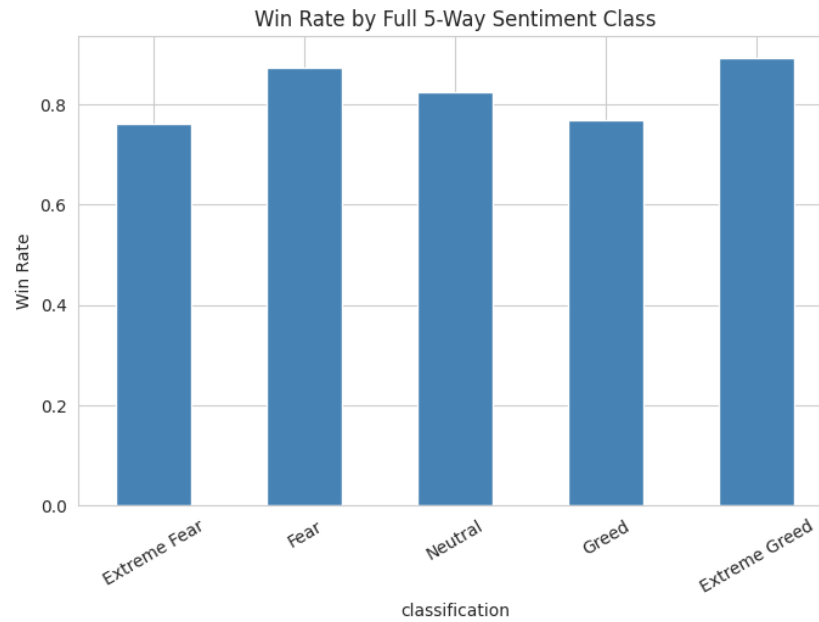


Figure 7. Win rate across the full five-way sentiment classification.

6.2 Closed PnL Distribution

Average closed PnL, split further by coin group, shows no consistent direction: for majors (BTC/ETH), average PnL is markedly higher during Fear; for other/altcoins, it's actually highest during Greed. This inconsistency undercuts the case for a robust sentiment-PnL relationship.

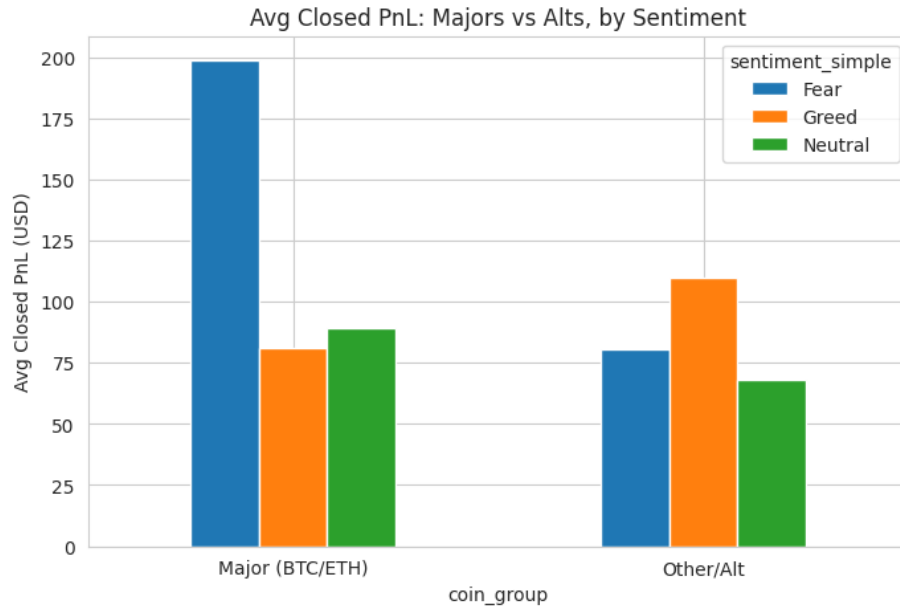


Figure 8. Average closed PnL (USD) by sentiment, split by major coins (BTC/ETH) vs. other/altcoins.

Formal testing confirms this: a Mann-Whitney U test comparing Fear vs. Greed closed PnL returns $p = 0.626$ (not significant), with a negligible rank-biserial effect size of 0.002. Similarly, the Pearson correlation between the raw sentiment value and daily PnL is $r = -0.083$ — a very weak negative relationship that does not indicate meaningful predictive power.

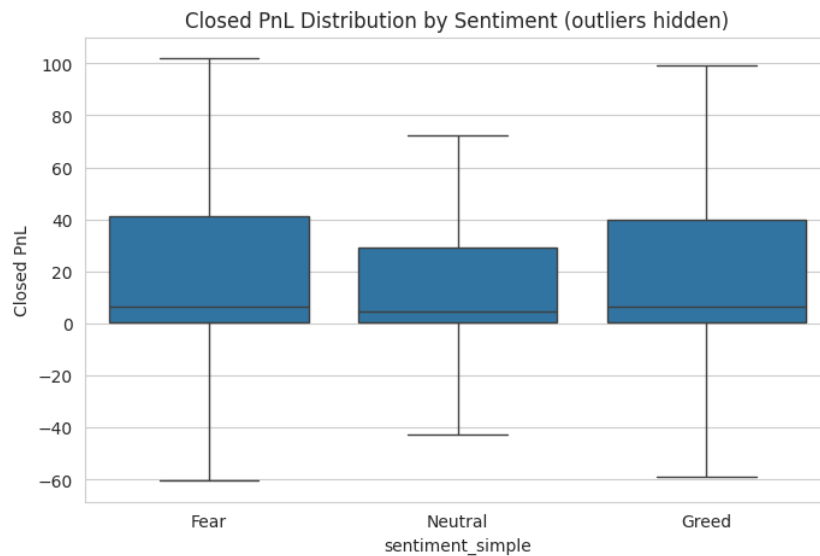


Figure 9. Closed PnL distribution by sentiment (boxplot, extreme outliers hidden).

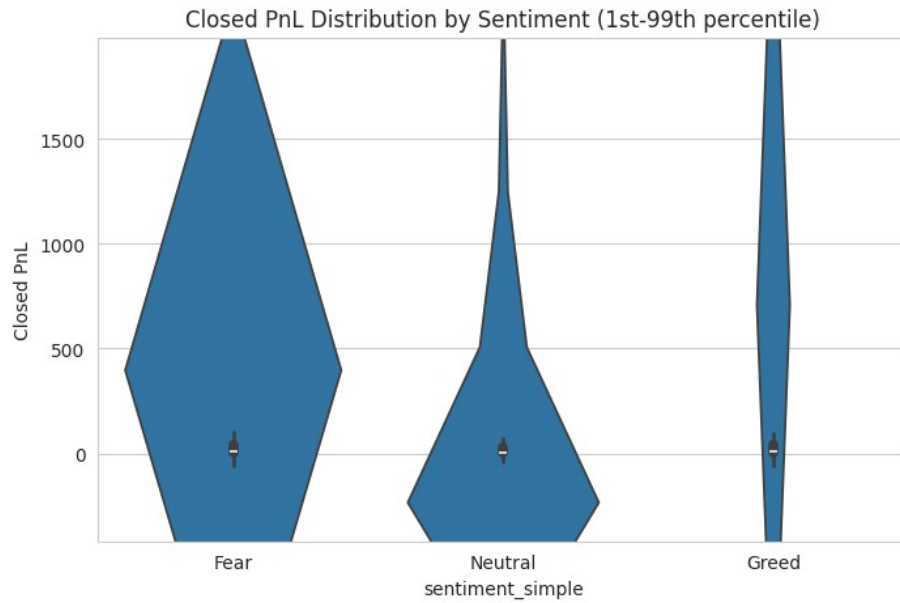


Figure 10. Closed PnL distribution by sentiment, 1st–99th percentile view (violin plot) — shapes are broadly similar across regimes.

6.3 Win Rate by Individual Account

The account-level heatmap makes clear that the single biggest source of variation in win rate is the trader, not the sentiment regime. Some accounts win consistently near 100% regardless of sentiment; others swing from 0% to 100% between sentiment buckets purely due to small sample sizes in that account/sentiment cell. This account-level variance is a key reason overall sentiment effects appear muted in aggregate.



Figure 11. Win rate by account and sentiment (green = higher win rate, red = lower).

7. Predictive Modeling

Two modeling tasks were run: (1) classifying whether an individual trade will win or lose, and (2) regressing the closed PnL value directly. Each task was tested with a simple linear/logistic baseline and a Random Forest.

Model	Task	Metric 1	Value 1	Metric 2	Value 2
Logistic Regression	Win / Loss	Accuracy	73.6%	AUC	0.703
Random Forest Classifier	Win / Loss	Accuracy	73.9%	AUC	0.529
Linear Regression	Closed PnL	RMSE	1,763.3	R ²	-0.262
Random Forest Regressor	Closed PnL	RMSE	1,664.0	R ²	-0.124

The classification models perform reasonably (Logistic Regression achieves 0.703 AUC), while the regression models both post negative R², meaning they perform worse than simply predicting the mean PnL — closed PnL at the individual trade level is highly noisy and not well explained by the available features.

7.1 Feature Importance

Feature importance from the Random Forest classifier shows a clear hierarchy: a trader's own rolling historical win rate is by far the strongest predictor of whether their next trade wins (importance 0.583) — more than double the next-strongest feature, the previous day's sentiment value (0.227). Trade-level factors like side, notional size, sentiment category, and coin group contribute comparatively little.

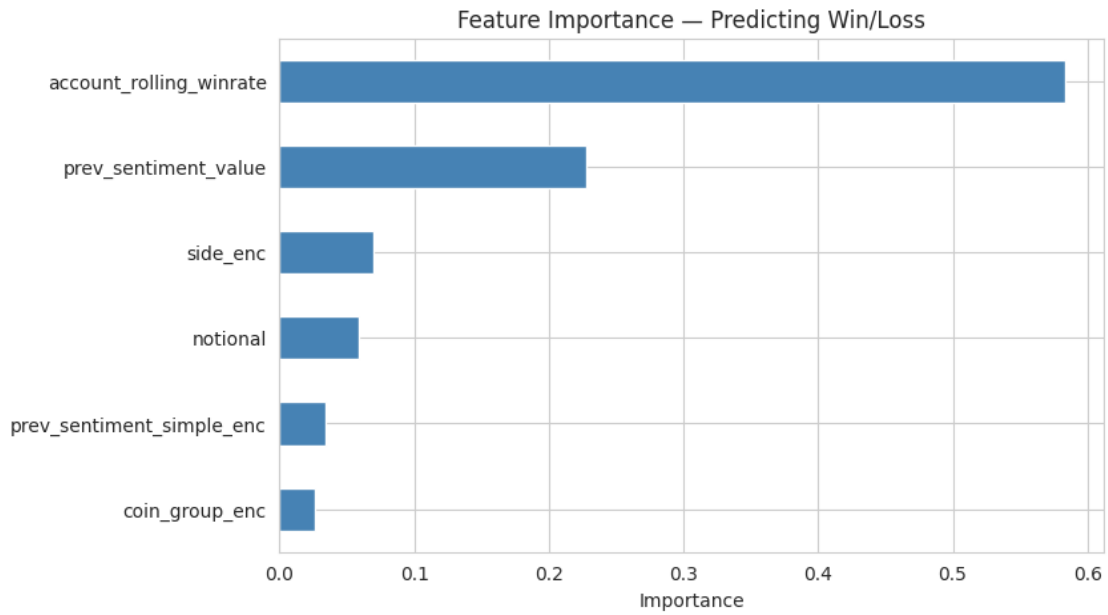


Figure 12. Feature importance for predicting win/loss (Random Forest Classifier).

This is the clearest finding in the entire analysis: individual trader skill and consistency outweigh market sentiment as a determinant of trade success — by a factor of roughly 2.5x in relative importance.

8. Key Findings — Summary

- Trading activity, volume, and participation all rise sharply during Fear and fall during Greed — sentiment strongly shapes how much traders trade.
- Directional bias shifts meaningfully with sentiment: traders go longer during Fear (58%) and shorter during Greed (only 33% long) — statistically significant (χ^2 $p < 0.00001$, Cramér's $V = 0.18$).
- Win rate differs only marginally across sentiment buckets (82.3%–84.4%) — a spread of about 2 percentage points.
- Closed PnL shows no statistically significant difference between Fear and Greed (Mann-Whitney $p = 0.626$); the sentiment value / daily PnL correlation is weak ($r = -0.083$).
- Classification models modestly predict win/loss (AUC up to 0.70); regression models fail to predict PnL magnitude (negative R^2 for both).
- A trader's own rolling win rate is, by far, the most important predictor of trade outcomes — more important than sentiment, trade size, side, or coin group combined.

9. Caveats & Limitations

- Only 32 unique accounts are represented — account-level win rate patterns may not generalize to a broader trader population.
- Sentiment is captured as a single daily value shared across all trades that day, which limits its granularity and predictive resolution.
- Negative R^2 in the PnL regression models indicates the available features do not meaningfully explain PnL magnitude — results should not be used to predict specific trade outcomes.
- These findings should be read as directional, exploratory hypotheses rather than validated trading rules.